Trending and emerging prospects of physics-based and ML-based **wildfire spread** models: A comprehensive **review**

H Singh, LM Ang, T Lewis, D Paudyal, M Acuna… - Journal of Forestry …, 2024 - Springer

… In addition, remote sensing data is used to create spatial **maps** encompassing components … Methods for literature **review** This **wildfire spread model review** analysed research published …

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Current approaches to modelling the **spread** of **wildland fire**: a **review**

GLW Perry - Progress in Physical Geography, 1998 - journals.sagepub.com

… for **wildland fire management** and how such models may be integrated within land … the success of fire **management**, **mapping** fire hazard for fire control planning and the production of …

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Machine learning and deep learning for **wildfire spread prediction**: A **review**

HS Andrianarivony, MA Akhloufi - Fire, 2024 - mdpi.com

… a systematic **review** of recent ML and DL techniques developed for **wildfire spread prediction**, … The **model** outputs a 64 × 64 binary fire **map**, where each pixel represents the likelihood of …

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Uncertainty and risk in **wildland fire management**: a **review**

MP Thompson, DE Calkin - Journal of environmental **management**, 2011 - Elsevier

… **review** the state of **wildfire** risk assessment and **management**… advances in **wildfire simulation** and geospatial **mapping** of … factors contributing to **wildfire** occurrence and **spread**. Managing …

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A Comprehensive **Review** of Empirical and Dynamic **Wildfire** Simulators and Machine Learning Techniques used for the **Prediction** of **Wildfire** in Australia

H Singh, LM Ang, D Paudyal, M Acuna… - Technology, Knowledge …, 2025 - Springer

… , fuel load and conditions, grass curing, slope, and the selected fire **spread model** to compute **wildfire spread** predictions and draw their perimeters on **maps** of the area of interest. …

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A comparative **review** on **wildfire** simulators

GD Papadopoulos, FN Pavlidou - IEEE systems Journal, 2011 - ieeexplore.ieee.org

… This project promises to develop more reliable **bushfire spread** simu… and plot the perimeter on a **map** of the area of interest. GIS … In this paper, a **review** on **wildfire simulation** models has …

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A **review** of machine learning applications in **wildfire** science and **management**

P Jain, SCP Coogan, SG Subramanian… - Environmental …, 2020 - cdnsciencepub.com

… For example, in the task of modeling **wildland fire spread**, the relevant physical properties … This was the only application of DL that we could find for fire-**susceptibility mapping**. The …

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Facing the **wildfire spread** risk challenge: where are we now and where are we going?

J Sun, W Qi, Y Huang, C Xu, W Yang - Fire, 2023 - mdpi.com

… Consequently, it is imperative to comprehensively **review** … the following results for **wildfire spread simulation**. In Figure 6, … the visualized **maps** of **wildfire spread behavior** after changing …

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A **review** of genetic algorithm approaches for **wildfire spread prediction** calibration

J Pereira, J Mendes, JSS Júnior, C Viegas, JR Paulo - Mathematics, 2022 - mdpi.com

… So, this work introduced a wind field **model** (WindNinja), being represented by a cell **map**, to … Then, having each individual's wind field, the corresponding fire **propagation map** is …

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A systematic **review** and bibliometric analysis of **wildland fire behavior** modeling

J Silva, J Marques, I Gonçalves, R Brito, S Teixeira… - Fluids, 2022 - mdpi.com

… percept to **model** the **spread** of **wildland fire**. Bakhshaii and Johnson [7], in a recent **review**, called these last … Fire danger **maps** can be developed based on probabilistic fire simulations. …

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A **review** of **wildland fire spread** modelling, 1990-present 3: Mathematical analogues and **simulation** models

AL Sullivan - arXiv preprint arXiv:0706.4130, 2007 - arxiv.org

… to plot the likely **spread** of the fire on a **map**, thereby putting the **prediction** in context with … of fire **spread simulation**. Because much of the development of the early **wildland fire** behaviour …

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Research Trends in **Wildland Fire prediction** amidst climate change: A Comprehensive bibliometric analysis

M Bao, J Liu, H Ren, S Liu, C Ren, C Chen, J Liu - Forests, 2024 - mdpi.com

… conduct a critical thematic **review** of **wildland fire prediction** through a … **maps**, we identified the current focal areas in **wildland fire** … cluster explores **wildfire behavior** and **spread simulation**, …

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Wildland surface fire **spread** modelling, 1990–2007. 3: **Simulation** and mathematical analogue models

AL Sullivan - International Journal of **Wildland Fire**, 2009 - CSIRO Publishing

… **survey** and précis all types of surface fire **spread** models … the likely **spread** of the fire on a **map**, thereby putting the **prediction** in … , and constituted the first form of fire **spread simulation**. …

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Application of deep learning in **forest fire prediction**: a systematic **review**

C Mambile, S Kaijage, J Leo - IEEE Access, 2024 - ieeexplore.ieee.org

… (1D and 2D) for **wildfire mapping** and **prediction** assessment, and the FARSITE simulated … poral convolutional neural network **model** for **wildfire spread prediction**. The daily ignition …

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Simulation of **wildfire** evacuation with dynamic factors and **model** composition

A Beloglazov, M Almashor, E Abebe, J Richter… - **Simulation** Modelling …, 2016 - Elsevier

… **simulation** systems are essential tools for such planning and decision making. During a **wildfire** … Our approach composes several **simulation** and modelling systems, including a **wildfire** …

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Mathematical models and calculation systems for the study of **wildland fire** behaviour

E Pastor, L Zárate, E Planas, J Arnaldos - Progress in Energy and …, 2003 - Elsevier

This is a **review** of the most important work in **wildland fire** mathematical modelling which … **wildland fire** models to be sorted. Surface fire **spread** models, crown fire initiation and **spread** …

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A **review** of recent advances in risk analysis for **wildfire management**

C Miller, AA Ager - International journal of **wildland fire**, 2013 - connectsci.au

… **review** recent advances in addressing temporal **dynamics** of fire risk and spatial optimisation of fuels **management** … These papers quantified and **mapped** one or more of three **wildfire** …

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Wildland fire spread modeling using convolutional neural networks

JL Hodges, BY Lattimer - Fire technology, 2019 - Springer

… Over 10,000 **model** predictions were made to determine burn **maps** in 6 h increments up to … The objective of this study was to develop a machine learning **wildland fire spread model** to …

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Wildfire prediction to inform fire **management**: statistical science challenges

SW Taylor, DG Woolford, CB Dean, DL Martell - 2013 - projecteuclid.org

… In this paper we **review** some of the models that have been … of a particular **spread model** against **wildfire** observations. A … currence were **forecast** and then were stochastically **mapped** to …

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A method for ensemble **wildland fire simulation**

MA Finney, IC Grenfell, CW McHugh, RC Seli… - … Modeling & Assessment, 2011 - Springer

… **simulation** system that accounts for uncertainty in long-range weather conditions and two-dimensional **wildland fire spread** … , digital fuel and vegetation **maps**, and enhanced methods for …

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